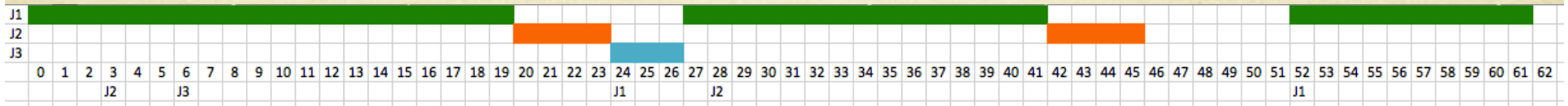


Scheduling policies for batch systems

First come first served (*FCFS*)

- ◆ Scheduler maintains *queue*
 - ◆ Incoming job added to end of queue
 - ◆ Job at head of queue chosen for execution
- ◆ Jobs scheduled in *non-preemptive* manner
 - ◆ Once job starts, it continues until it blocks/terminates/yields
 - ◆ Blocked job removed from queue
 - ◆ Upon unblocking, job added to end of queue
- ◆ *Pros*: very simple
- ◆ *Cons*: short jobs may have long wait time

Job	Arrival time	CPU burst	I/O time
J1	0	20	
			4
		15	
			10
		10	
J2	3	4	
			4
		4	
J3	6	3	

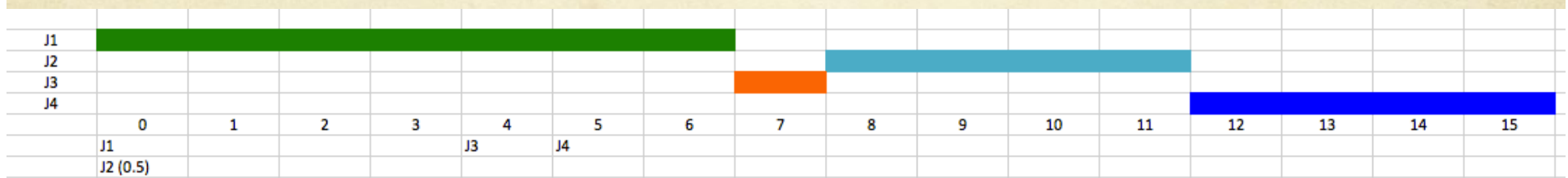


Shortest-Job-First (*SJF*)

- ◆ Scheduler chooses *shortest* job for execution at any given time
- ◆ Two *versions* of SJF may be used
 - ◆ Version 1: *Non-preemptive*
 - ◆ Once job is scheduled, it cannot be *preempted* until it completes CPU burst
 - ◆ Version 2: *Preemptive*
 - ◆ New job with *shorter* CPU burst than *remaining time* of curr job arrives
 - ◆ Preempt current job & schedule new job
 - ◆ A.k.a. Shortest-Remaining-Time-First (*SRTF*)
- ◆ *Pros*: Good for short jobs
- ◆ *Cons*:
 - ◆ Needs *a-priori* knowledge of *execution times* of CPU bursts
 - ◆ Long jobs may *starve*

Non-preemptive SJF example

Job	Arrival time	CPU burst
J1	0	7
J2	0.5	4
J3	4	1
J4	5	4



Preemptive SJF (SRTF) example

Job	Arrival time	CPU burst
J1	0	7
J2	2	4
J3	4	1
J4	5	4

